



Solution - ss23

Kognitive Systeme (Technische Universität München)



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- Sign in the corresponding signature field.

Cognitive Systems

Exam: IN2222 / Endterm
Examiner: Prof. Knoll

Date: Thursday 3rd August, 2023
Time: 13:30 – 14:45

	P 1	P 2	P 3	P 4	P 5	P 6
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Working instructions

- This exam consists of **20 pages** with a total of **6 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 150 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources: None
- Write your name and matriculation number on the top of **every** page.
- **Give concise answers using appropriate terminology.**
- You may only answer in English.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.
- You have **75 minutes** working time

Left room from _____ to _____ / Early submission at _____

Problem 1 Cognition and Cognitive Science (29 credits)

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a) Explain the difference between cognition and intelligence.

Cognition is a global process at the system level (2) that integrates (2) many different processing modalities.

Special cognitive skills (2) such as intelligence, learning, memory, interaction, etc. are constituents and synergies of a cognitive system.

C02-1 S18

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b) State briefly the two questions that are answered by the *proximate* and the *ultimate* explanation of a cognitive skill.

Proximate explanation: How is a certain behavior implemented in a cognitive system? (2)

Ultimate explanation: Why does a cognitive system exhibit a certain behavior? (2)

C03-1 S19

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c) Provide a brief definition of joint action.

Humans and robots cooperate (1) (mentally and physically) (1) in a joint workspace by synchronising (1) their actions and mental states.

C01-6 S2

d) Perception is an important stage in the cycle of cognitive processing.
Draw the whole cycle with its other processing stages including all subdivisions as defined in the lecture.

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1 credit for each correct stage and arrow

... -> Perception -> Cognition (-> Anticipation -> Assimilation -> Adaption ->) -> Action -> Perception
C02-1 S20

e) Large language models mark a major advance in natural language processing, which is a central cognitive skill and an important component of joint action.
Name three of the key innovations that enabled the development of ChatGPT.

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- Training of a generative language model (1)
- A new neural network architecture for text called the Transformer (1)
- A new training procedure for learning dialogues (1)
- Massively large scaling of the model (1)

C01-5 S20

f) State-of-the-art large language models such as ChatGPT can produce engaging dialogues that may, under some circumstances, even pass the Turing test. This raises the general question if the Turing test is an adequate means for assessing the cognitive capabilities of artificial systems.
State the name of a thought experiment that was formulated as a counterargument to the Turing test.
Provide a detailed description of the experiment.

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Name: Chinese Room Argument (1)

- A person with no knowledge of the Chinese language sits in a room (1).
- The room contains an archive with rules that map Chinese questions to responses (1).
- People can submit written questions through a slot in the room (1).
- The person answers these questions using the rules like a computer program (1).
- Even though the answers are correct (1), the person does not understand the meaning (1).

C02-2 S27

Problem 2 Cognitive Architectures and Neurorobotics (21 credits)

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a) Name three types of cognitive architectures.

Symbolic, Emergent, Hybrid

E09 S12

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b) Describe two advantages of using a cognitive architecture like ACT-R instead of a standard programming language.

- They advance in the understanding of human cognition.
- They are generic and therefore can be used in different domains.
- They are unified models that coherently integrate different aspects of cognition.
- They are executable implementations of theories about cognition.
- They are deep explanations of cognitive processing.

2 points per correct answer. 4 max

E09 S9

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c) Provide the names of the four main modules of ACT-R.

Intention module (1), declarative module (1), visual module (1), manual module (1)

- Caution: action != manual.

- Procedural module can also be considered as one of the main modules.

E09 S24

d) You are asked to contribute to an ACT-R implementation of a calculator.
Define a chunk type "multiplication" and describe the equation $11 \cdot 3 = 33$ as declarative knowledge in ACT-R syntax.

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Define chunk type (1).

```
(chunk-type multiplication factor1 factor2 product)
```

Fill chunk (correct syntax with a meaningful name (1), correct type and variables (1)) and add it to declarative module (1).

```
(add-dm
  (Multiply113:
    ISA multiplication
    factor1 11
    factor2 3
    product 33))
```

E09 S48

e) Describe the role of the Transfer Function (TF) in the Neurorobotics Platform (NRP).
Name four different tools that can be interactively used with the Transfer Function.

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It connects and synchronizes brain models to robot models. (2)

- ROS/Gazebo
- OpenSim
- NEST
- Neuromorphic Hardware
- Custom Brain Models with python Libraries such as Tensorflow, Pytorch
- Physical Robot in real world

(one point per item, max four points)

E03 S15-18

Problem 3 Human Brain (20 credits)

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a) Explain whether it is true that only ten percent of the brain is used.

Not true, every part of the brain is used but not all parts are used at once (2).
Reference: C04-1 18

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b) Name two streams of visual processing in the human brain and briefly outline the type of information computed along each of the streams.

Dorsal stream (1):
- The dorsal pathway of visual processing determines an object's spatial location ("where?"). In a more recent view, it is assumed to determine how motor actions are carried out ("how?"). (1)
Ventral stream (1):
- The ventral pathway of visual processing determines an objects identity ("what?"). (1)
Reference: C05-2 12

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c) In 1961 D. H. Hubel and T. N. Wiesel discovered two types of cells in the visual cortex, namely simple cells and complex cells.
Name the two important attributes of simple cells, i.e. name the attributes of features that do activate them.

- basic shapes, such as points or lines or edges
- correctly oriented
- correctly localized

One of the above solutions gives two points.
Reference: C05-2 13

d) In 1943 Warren McCulloch and Walter Pitts proposed an artificial neuron model. Around ten years later Alan Hodgkin and Andrew Huxley described their biological neuron model. Name two key differences between analog neuron models (artificial neuron models) and spiking neuron models (biological neuron models).

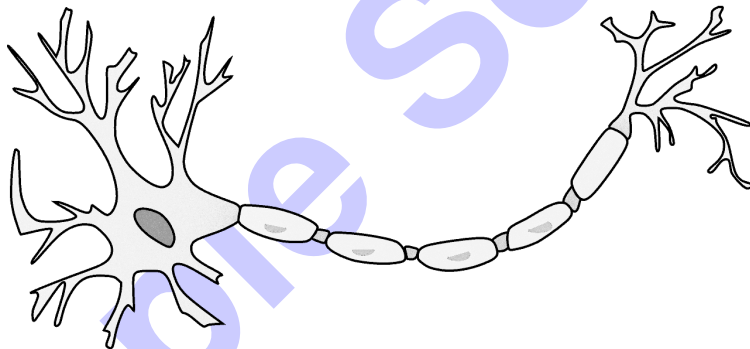
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- Analog neuron models are functions that are not able to reproduce the temporal dynamics of biological neurons.
- Analog neurons do not generate spikes.
- dynamical system vs activation function
- digital spikes vs analog values
- rate vs real numbers
- high complexity vs low complexity
- high biological plausibility vs low

2 points per item, max 4 points
(Reference: E04, Slide 38)

e) Annotate the names of the three main components of a biological neuron in the given sketch below. Describe each component briefly.

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- Soma (1): cell body containing the nucleus (1)
- Dendrites (1): receive electrical impulses (1)
- Axon (1): conducts electrical impulses (1)

Reference: E04 S. 9

Problem 4 Biological Neurons and Spiking Neural Networks (28 credits)

The membrane potential of a simplified Leaky Integrate and Fire (LIF) neuron is described as a single differential equation

$$C \frac{dv}{dt} = I(t) - \frac{v(t)}{R}, \quad (4.1)$$

with membrane capacity C , membrane resistance R , and input current $I(t)$. Spikes of LIF neurons are generated if the membrane potential reaches a threshold value v_{thresh} . After the occurrence of a spike the membrane potential is immediately set to $v_{\text{reset}} = v(0)$, i.e., we have no refractory period.

a) Name two essential features of neural processing that are captured by the *leaky integrate-and-fire* model.

natural permeability, passive ion channels, concentration gradient, membrane as capacitor

Reference: Ex05 - problem sheet

b) Solve the differential equation 4.1 for the case of a constant input current $I(t) = I$, given an initial membrane potential $v(0) = 0$ at initial time $t_0 = 0$.

- Separation of variables (2 points)

$$\int dv \frac{1}{v - RI} = \int dt \frac{-1}{R \cdot C}$$

- Correct integration (fraction to log) and limits (4 points)

$$\begin{aligned} \int_{v_0=0}^v dv \frac{1}{v - RI} &= \int_{t_0=0}^t dt \frac{-1}{RC} \\ \Rightarrow \ln\left(\frac{v - RI}{-RI}\right) &= -\frac{t}{RC} \end{aligned}$$

- Correct result (2 point)

$$\Rightarrow v(t) = (-RI) \cdot \exp\left(-\frac{t}{RC}\right) + RI$$

Reference: Ex05 - problem sheet

c) A futuristic company builds a new humanoid robot and is asking for your help. The robot should be equipped with neuromorphic sensors that encode the information with spikes. They want to ensure fast reaction times for their robot and ask you which encoding technique they should use.

- Describe the two existing encoding techniques.
- State the encoding technique that is suited for the given problem.

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Rate coding: The information is carried in the averaged spike rate within a certain time window (e.g. muscle excitation). (2)

Time-to-First Spike Coding: The information is encoded in the time of emission of the first spike after a reference point (e.g. onset of a stimulus). (2)

Time Coding for the given problem (1)

Reference: EX06, slide 8

d) The Hebbian theory can be summarized as "Cells that fire together wire together".

Name the three fundamental properties for a Hebbian learning algorithm to be biologically plausible and stable over an undetermined period of time.

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Saturation (1), locality (1), or competition/selectivity (1)

Reference: EX05 - slide 11

e) In the context of Hebbian learning, describe what "saturation" means, and state the problem when a Hebbian learning algorithm does not possess this feature. You might draw a graph for illustrating the problem.

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If a learning rule has saturation, the growth of the weights is limited to certain soft or hard boundaries. (Can also be explained with a graph) (2)

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- f) Equation (4.2) is a basic form of Hebbian learning.
 Modify the equation (or add a second one) in order to add saturation to the learning rule.

$$w_t = w_{t-1} + \frac{\Delta t}{\tau_w} v \cdot u \tag{4.2}$$

A solution is to set a hard boundary that cannot be reached by the weights (2 points are only added if the equation is missing)

$$w_t = \min \left\{ w_{t-1} + \frac{\Delta t}{\tau_w} v \cdot u, w_{max} \right\}$$

(4 points)

Reference: EX05 - slide 11

The brain changes over time and is able to adapt itself to new situations. In this context, different types of neuroplasticity have been discussed in the course.

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- g) Explain the term *structural plasticity*, i.e. the type of changes performed on the structure and function of the nervous system.

structural plasticity: physical creation and deletion of synapses (3 points)

Reference: Ex05 - slide 31

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- h) State whether or not the structural plasticity plays a role when it comes to learning.

yes (1 point)

Reference: Ex05 - slide 31

Problem 5 Machine Learning (22 credits)

Suppose you have collected an image dataset $D = \{\mathbf{x}, \mathbf{y}\}$ that includes photos of 1000 different people's faces $\mathbf{x}$, and relevant information $\mathbf{y}$ (e.g., age, gender, ...) of each image. In general, samples $(\mathbf{x}, \mathbf{y})$ drawn from D are distributed according to probability $P(\mathbf{x}, \mathbf{y})$.

Your colleagues want to train different models from machine learning on this dataset.

a) Colleague A wants to train a model that takes one image as input, and predicts the gender of the person on the image.

State the type of the model that colleague A is training. Write down the conditional probability that this model is relying on.

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- Discriminative model (or Classification model) **1p.**
- Based on the posterior probability $P(\mathbf{y}|\mathbf{x})$ **1p.**

→E02 p37

b) After training, colleague A finds that the model performs well on the training dataset, but performs badly on the test dataset.

State the problem that A's model is suffering from. Name three methods to avoid such problem.

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Overfitting **1p.**

Any three of them **3p:** Regularization / dropout, More Training Data, Dataset Augmentation, Early stopping, Simpler models →E02 p30 and p42

c) Colleague A adjusts the training following methods in the last question, and finally gets three models each with 1000, 2000 and 2500 parameters. These models have similar performances on the test dataset.

State the model that A should choose from the above three. Name and explain the principle that supports your choice.

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- A should choose the model with 1000/the least parameters. **1p.**
- Occam's Razor **1p:** Of two competing theories, the simpler explanation of an entity is to be preferred **1p.**

→E02 p32

d) Colleague B wants to train a model that takes a desired gender and age as input, and generates the corresponding face image.

State the type of the model that colleague A is training.

Write down the conditional probability that this model is relying on.

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- Generative model **1p.**
- Based on prior probability $P(\mathbf{x}|\mathbf{y})$ (or equivalently $P(\mathbf{x}, \mathbf{y})$) **1p.**

→E02 p37-38

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e) Name and explain three different learning paradigms.

1p name, 1p explanation, 6p max

- Unsupervised learning **1p**: learning of statistical regularities in the input data **1p**
- Reinforcement learning **1p**: learning which action to chose in a certain situation in order to maximize an external reward signal through interaction with the environment **1p**
- Supervised learning **1p**: learn a specific mapping from input to output signals based on explicit instructions by a teacher **1p**
- Semi-Supervised learning **1p**: mixed dataset (labeled / unlabeled data) **1p**

→E01 p24

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f) Describe the cross-validation technique.

Explain how to measure the overall performance when using this technique.

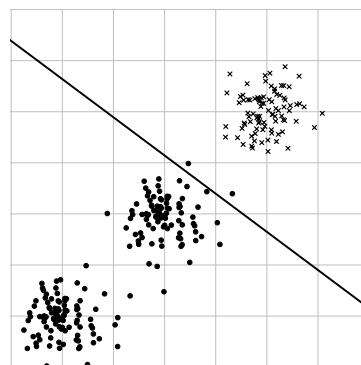
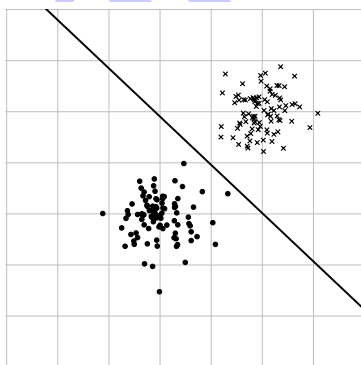
- The dataset is partitioned into k subsets and learned in k iterations **1p**
- In every iteration, a different subset is selected as validation set **1p**
- The overall performance corresponds to the averaged performances of the k iterations **1p**

→E01-2, slide 41

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g) Each of the two graphs below depicts a dataset with two classes of samples and a decision boundary learned by a linear classifier that was trained to minimize the least squares error.

Explain a major drawback of this classification model based on the two graphs, and provide the full name of a non-parametric classification method that does not suffer from this issue.



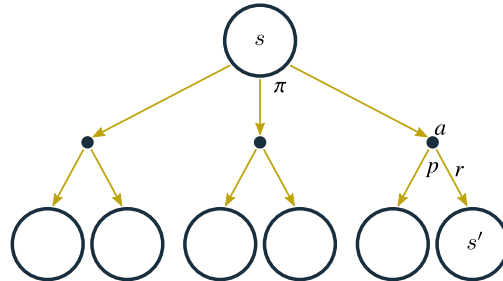
- This shows that least squares classification is sensitive to outliers **1p**.
- Support vector machines are non-parametric models that do not suffer from this issue. **1p**

→E02 69

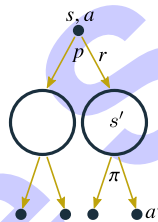
Problem 6 Reinforcement Learning (30 credits)

a) The Bellman equations define a recursive relationship between values of consecutive states or state-action pairs. Consider the following Bellman equation written for a state-value function with its corresponding backup diagram.

$$v_{\pi}(s) \doteq \mathbb{E}_{\pi}[G_t | S_t = s] = \sum_a \pi(a|s) \sum_{s', r} p(s', r | s, a) [r + \gamma v_{\pi}(s')]$$



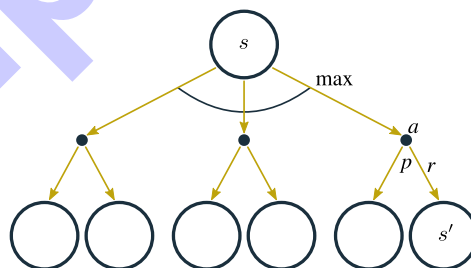
Backup diagram for an action-value function $q_{\pi}(s, a)$ (2) C06 S 2:9



Bellman optimality equation for the optimal state-value function:

$$v_*(s) = \max_a \sum_{s', r} p(s', r | s, a) [r + \gamma v_*(s')] \quad (1) \text{ for } \max_a \text{ and } (1) \text{ for recursive } v_*$$

Backup diagram for the Bellman optimality equation (1) C06 S 2:11



(1) point by answering any of the following algorithms. Mentioning only DP or TD is also acceptable.

- (Dynamic Programming) Value Iteration C06 S 3:12
- (Temporal Difference) Q-learning C06 S 5:11

The python function below is part of a reinforcement learning agent and is called whenever the agent wants to take an action for execution on the environment.

```
def policy(state):
    if np.random.rand() <= epsilon:
        action = np.random.randint(action_size)
    else:
        values = model.predict(state)
        action = np.argmax(values[0])
    return action
```

- 0 ☐ b) A behavior policy is a policy that is used to select an action based on received observations from the
1 ☐ environment. A target policy is a policy that the agent uses for improving the estimated value functions
according to expected returns.
State if the above function *policy* implements a behavior policy or a target policy.

the behavior policy (1p)

E08 S10

- 0 ☐ c) State the type of action space (continuous, discrete or both) that can be handled by the function *policy*.
1 ☐

discrete action space (1p)

E08 S10

- 0 ☐ d) Assume that in the function *policy*, the object *model* is a feed-forward neural network which approximates
1 ☐ the value function.
State if the approximated value function is a state-value function $v_{\pi}(s)$ or an action-value function $q_{\pi}(s, a)$.

action-value function $q_{\pi}(s, a)$ (1p)

E08 S10

- 0 ☐ e) The implemented *if-else* statement in the function *policy* defines a policy for balancing between exploration
1 ☐ and exploitation steps.
2 ☐ State how this specific policy is called. Write down the equations that formulate $\pi(a|S_t)$ for the implemented
3 ☐ policy.

ϵ -greedy policy (1p)

$$\pi(a|S_t) \leftarrow \begin{cases} 1 - \epsilon + \epsilon/|A(S_t)| & \text{if } a = A^* \text{ (1p)} \\ \epsilon/|A(S_t)| & \text{if } a \neq A^* \text{ (1p)} \end{cases}$$

E08 S10

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f) Consider an environment which lets the agent to execute one single action out of four possible actions at every time step (i.e. $A(S_t) \in \{A_0, A_1, A_2, A_3\}; \forall t$). Assume that the agent is relying on the implemented *policy* function to decide which action to command.

- Write down the general form of the equations formulating $\pi(a|S_t)$ for this specific agent.
- Compute the probability to choose a non-optimal action at every time step.

Consider to have an extra fifth action A_4 that can be executed at every time step.

i.e. $A(S_t) \in \{A_0, A_1, A_2, A_3\}; \forall t$ is changed to $A(S_t) \in \{A_0, A_1, A_2, A_3, A_4\}; \forall t$.

- Compute the decrease of probability of selecting the optimal action at every time step by this change.

$$\pi(a|S_t) \leftarrow \begin{cases} 1 - 3\epsilon/4 & \text{if } a = A^* (1p) \\ \epsilon/4 & \text{if } a \neq A^* (1p) \end{cases}; a \in \{A_0, A_1, A_2, A_3\}$$

The probability to choose non-optimal (non-greedy) actions is $3\epsilon/4$. (1p)

The probability to choose the optimal (greedy) action is $1 - 3\epsilon/4$ for the case of having four actions and $1 - 4\epsilon/5$ for the case of having five actions.

Therefore, we have $(1 - 3\epsilon/4) - (1 - 4\epsilon/5) = \epsilon/20$. (1p)

E08 S10

g) Fill in the blanks: Classical and operant conditioning are two basic _____ learning processes that link sensory stimuli to corresponding _____.

State the main differences between the two forms of conditioning by providing detailed description of each.

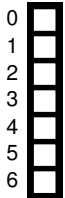
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Classical and operant conditioning are two basic **associative** (1) learning processes that link sensory stimuli to corresponding **actions** (1).

In classical conditioning, a subject learns the relationship between an initially neutral conditioned stimulus (CS) and an unconditioned stimulus (US) (1) that reflexively produces a conditioned response (CR) (1). When the subject is repeatedly presented the CS and US together, the CS will finally produce the CR even in the absence of the US. (1)

In operant conditioning, a subject learns the relationship between a stimulus and its behavior (1). A stimulus is only presented in response to a certain action of the subject (1) and serves as a reinforcer that increases or decreases the probability of that action. (1)

C06 S 1:7



h) Based on the reward hypothesis, the return G_t is the main term that a reinforcement learning agent should maximize in expectation. In the simplest case, the return is defined directly as the sum of all the future rewards $G_t := R_{t+1} + R_{t+2} + R_{t+3} + \dots + R_T$. In the case of *episodic* tasks, T is the episode length and is a finite integer number. However, in the case of *continuing* tasks, there is no episode termination and T is infinite.

- State the mathematical problem that the above definition of return may cause in *continuing* tasks.
- Provide a modified definition of return that solves such problem and state the resulting equation.
- Describe how the modified definition can let the agent to change from being farsighted to shortsighted.

1. The simple formulation of return is problematic for continuing tasks because the final time step would be $T = \infty$, and **the return**, which is what we are trying to maximize, **could easily be infinite** (2).
2. The agent should try to select actions so that the sum of the **discounted** rewards it receives over the future is maximized (1).

$$G_t := R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots + \gamma^{T-t-1} R_T \quad (1).$$

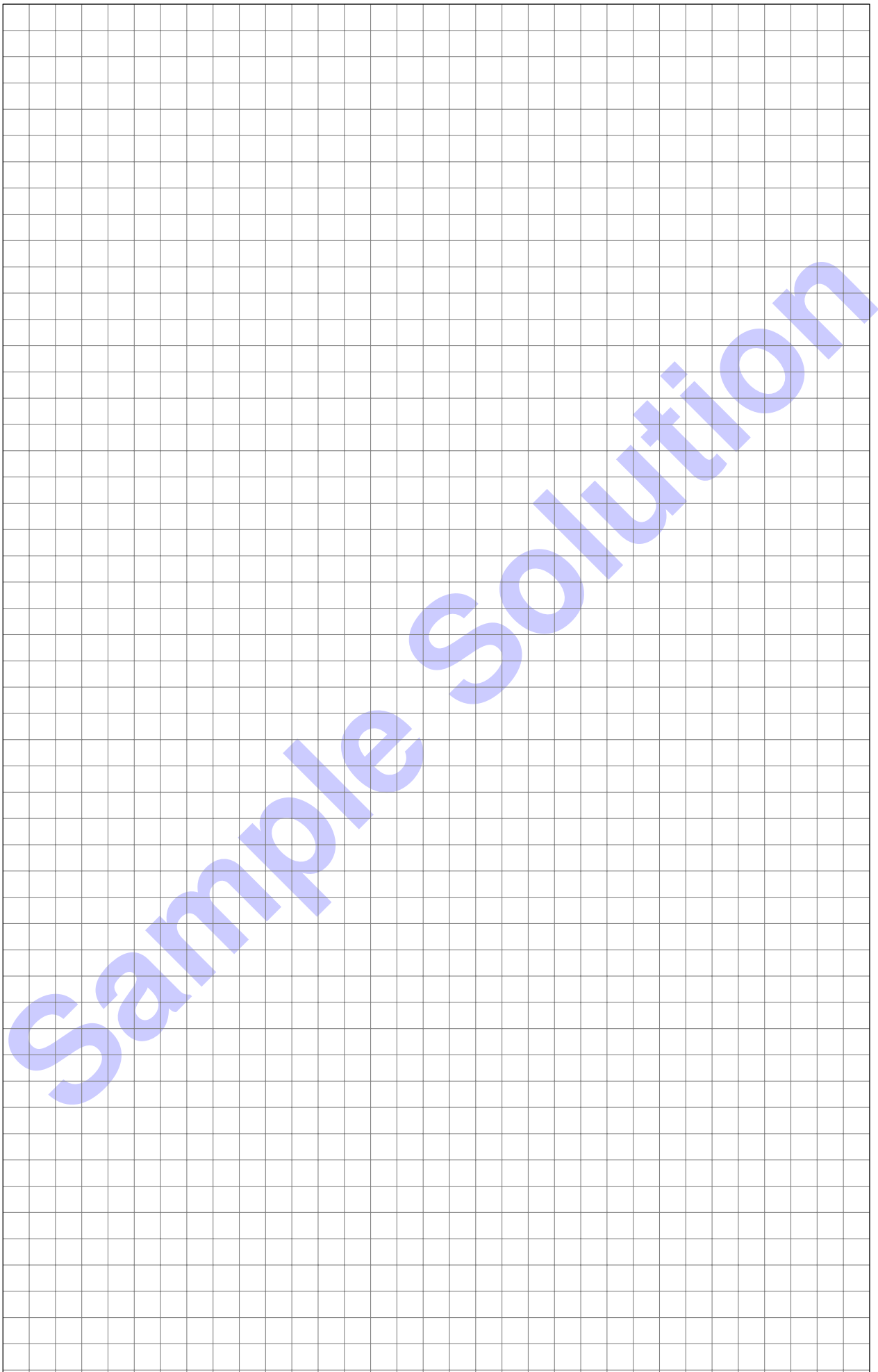
3. As γ approaches 0 (1), the return objective takes immediate rewards into account more strongly than the far future rewards (1); the agent becomes more shortsighted.

C06 S 1:18

Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

Sample Solution

Sample Solution



Sample Solution